

Identifiability and Early-Warning Reliability of Cusp Catastrophe Dynamics in Wearable Electrodermal Recordings

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Abstract—Sensory and executive overload is usually modelled descriptively: states are labelled, transitions counted, delayed recovery reported as an asymmetry. Such models cannot be falsified. We derive overload instead from the geometry of a cusp catastrophe. A latent load coordinate relaxes in a quartic potential whose asymmetry is set by demand and whose bistability is set by a slow recovery-debt variable; functional states become the metastable basins of that potential, and the transition kernel follows from Kramers escape rates. The formulation fixes two exponents in advance: hysteresis width scales as the $3/2$ power of the splitting factor, and near the fold variance and log-autocorrelation scale as $\mp 1/2$ in distance to the tipping point. Estimation is exact: conditional on the slow variable the drift is linear in a reparameterised coefficient vector. We report three findings, two about method. First, on near-unit-root series like tonic electrodermal activity the nominal test for the cubic term rejects 42% of random walks rather than 5%. Second, the rolling-window indicators the early-warning literature relies on return exponents of the wrong sign on data simulated from a model that genuinely contains a fold, in all twelve configurations tested, because the relaxation time outgrows the window exactly where the effect is largest; we withdraw our own early-warning test as uninterpretable. Third, across 147 recordings from 40 people, no cusp signature exceeds the calibrated null, which power bounds to excluding strong and moderate dynamics only. The work is motivated by autistic adults; none are analysed here.

Index Terms—Critical transitions, cusp catastrophe, hysteresis, early-warning signals, electrodermal activity, wearable sensing, state-transition models, sensory overload, executive function.

I. Introduction

Autistic adults frequently describe functional breakdown under combined sensory and cognitive demand as something that arrives suddenly and departs slowly [1]. Qualitative accounts are consistent on two points that any adequate model must reproduce: the onset is abrupt rather than proportional, and recovery outlasts the removal of the load that caused it. Empirical work supports the coupling that would produce this, with sensory atypicalities and executive difficulty repeatedly found to co-vary in adult samples [2], [3], [4].

Formal models have not kept pace. Computational accounts of autism have been dominated by Bayesian and predictive-coding framings [5], whose empirical support a ten-year review found mixed and methodologically inconsistent [6], and which are in any case formulated at the group level over static perceptual inference rather than over the minute-scale functional dynamics at issue. The most recent dynamical treatment of autism symptom trajectories is explicitly theoretical, deriving parameter ranges from the literature and collecting no empirical data [7]. Meanwhile the applied literature on state-transition modelling typically assigns discrete states from experimental condition labels and estimates a transition matrix by counting. That procedure is close to circular: the recovered matrix largely re-describes the block structure of the protocol, and it makes no prediction that could turn out to be wrong.

The shift this paper makes

We do not posit states and fit transitions between them. We posit a one-dimensional stochastic relaxation in a cusp potential and let everything else follow. The load coordinate x_t sits in $V(x; a, b) = \frac{1}{4}x^4 - \frac{1}{2}ax^2 - bx$, where the normal factor b_t carries instantaneous demand and the splitting factor a_t carries depleted regulatory capacity. The substantive claim is that regulatory capacity does not add to load—it changes the geometry. Below a critical a , load tracks demand smoothly and returns; above it a second attractor appears and the system acquires a fold, a hysteresis loop, and critical slowing down. Abrupt onset and slow recovery stop being observations to be reported and become consequences to be checked.

Because the geometry is closed-form, the consequences are quantitative. The hysteresis width is not merely positive but proportional to $a^{3/2}$; the variance of the load near the fold does not merely rise but rises as $\mu^{-1/2}$ in the distance μ to the tipping point. These are point hypotheses about exponents, and they can fail.

We present a cusp-catastrophe formulation of sensory-executive overload together with an identifiability analysis that bounds what any such analysis can recover from wearable data, and a ground-truth test of the early-warning estimators the formulation would otherwise be checked with.

- 1) A cusp-catastrophe formulation of sensory-executive overload (Sec. III) in which the discrete functional states are the metastable basins of a fitted potential, and the 5×5 transition kernel is generated by Kramers escape rates from nine interpretable parameters rather than by twenty counted frequencies. The states stuck and recovering, which qualitative work describes but no model defines, acquire exact geometric definitions.
- 2) Two closed-form scaling laws — hysteresis width $\propto a^{3/2}$ (Prop. 1) and early-warning exponents $-1/2$ and $+1/2$ (Prop. 2) — stated before estimation and tested as point hypotheses with bootstrap confidence intervals.
- 3) An exact estimator. Conditional on the slow variable the drift is linear in a reparameterised coefficient vector, so fitting reduces to profiled ordinary least squares over a single scalar: no bounds, no starting values, no local optima (Sec. IV-C).
- 4) A quantified warning about inference in this setting. On near-unit-root physiological series the nominal test for the governing cubic coefficient rejects 42% of random walks rather than 5%. Persistent series look bimodal and prefer flat-bottomed potentials simply by wandering, so matched unit-root surrogates, not nominal distributions and not the nested monostable restriction, must be the reference (Sec. IV-E).
- 5) A ground-truth validation of the early-warning toolkit itself, which it fails. Rolling-window autocorrelation and variance do not recover the predicted exponents on series simulated from a model that genuinely contains a fold; they return the wrong sign, at every window length and detrending choice tested, because the relaxation time diverges past the window exactly where the effect is strongest (Sec. V-F). We accordingly withdraw our own early-warning test rather than report an uninterpretable number.
- 6) A bounded null. Across 147 recordings from 40 people in three open corpora forming a gradient from fully controlled to fully naturalistic, the cusp signature is not detected above the calibrated null — including a case where the marginal distribution is decisively bimodal while the dynamics are not bistable, which is precisely the configuration that misleads a study reporting only distributional evidence (Sec. V). Of the five pre-stated predictions only one proves evaluable with these data and

instruments; we say which, and why, rather than reporting four numbers that would look like tests.

We report the null as the result rather than searching for a subgroup or a transformation in which the predictions survive. The predictions were stated before estimation, and the estimator’s power to detect the effect was established before the data were fitted; retrofitting either would forfeit exactly the property that makes the formulation worth having.

A statement about scope

The research programme this work belongs to targets autistic adults. None of the three corpora analysed here contains autistic participants, and no claim about autistic adults is made from them. Systematic reviews confirm that no public wearable dataset of autistic adults exists at the temporal density this model requires [8], [9]; the closest candidate [10] has released pre-registration materials but not raw data, and no autism-specific dense series appears in the open experience-sampling index [11]. Section VIII specifies the pre-registered protocol under which the predictions would transfer, and states in advance what would falsify them. We consider it better to publish a mechanism with an honest scope than to title a paper after a population it does not measure.

What the general stress corpora can settle is not whether autistic overload is a cusp, but whether the machinery could establish that it is. They carry the same sensor, the same sampling rate and the same recording lengths the intended study would, so the identifiability bound in Sec. V-A and the estimator failure in Sec. V transfer to that study directly. No claim about the population transfers, and none is made.

II. Related Work

A. Sensory-executive coupling in autistic adults

Meta-analytic evidence establishes executive difficulty as a replicable feature of autism [3], [4], and adult cohorts couple it to sensory atypicality with large effect sizes [2], [12]. Qualitative work is the most directly relevant to the present model: MacLennan et al. [1] describe overload not as an instantaneous event but as a staged cascade ending in shutdown, with recovery extending well past the removal of the trigger. That description is a qualitative statement of hysteresis, and it is what motivates modelling capacity as a bifurcation parameter rather than as an additive term.

B. Computational models of autism

Bayesian and predictive-coding accounts [5], [13] explain perceptual atypicality elegantly but are static and group-level; the supporting evidence is inconsistent across paradigms [6]. Adamou et al. [7] give continuous differential equations for symptom-domain trajectories across the lifespan, deriving parameters from theory and collecting no data. The present work differs on all three axes: discrete

metastable states rather than continuous symptom scores, minute-scale functional dynamics rather than lifespan trajectories, and parameters estimated from physiological time series rather than assumed.

C. Catastrophe theory in behavioural science, and its bad reputation

The cusp is the canonical two-control-parameter singularity [14], [15] and has an established behavioural-science lineage [16], most rigorously in van der Maas and Molenaar’s treatment of stagewise cognitive development [17] and in the estimation machinery of Grasman et al. [18]. That literature fits the cusp to cross-sectional response distributions. The complementary tradition of recovering latent discrete states from behavioural time series by hidden Markov modelling [19], [20] is closer to our data setting but assumes the states rather than deriving them from a potential. We use it dynamically: the potential governs a time series, and the catastrophe flags are tested as temporal scaling laws rather than as distributional shape statistics.

We should be direct about why that distinction matters. Applied catastrophe theory acquired a poor reputation for good reason: Zahler and Sussmann [21], [22] showed that many 1970s applications identified qualitative resemblances between data and cusp cartoons, chose control parameters post hoc, and stated no prediction that observation could contradict. Any sufficiently flexible geometry can be made to resemble anything. That critique applies in full force to a paper that merely asserts overload “is” a cusp.

Our defence is not that the critique has aged out; it is that this paper is constructed to be vulnerable to it. The control parameters are fixed in advance by measurement rather than chosen to fit. The predictions are exponents, $3/2$ and $\mp 1/2$, stated before estimation. And the null against which they are tested is calibrated by simulation rather than assumed. A qualitative resemblance argument cannot fail; an exponent with a bootstrap confidence interval can, and in Sec. V it does.

D. Critical transitions and early-warning signals

Approaching a fold, the leading eigenvalue of the linearised dynamics goes to zero, so perturbations decay ever more slowly [23]; the resulting rise in autocorrelation and variance is the standard early-warning indicator [24], [25], [26], and has been applied to depression [27], [28], [29] and reviewed critically for psychopathology [30]. The sceptical literature is equally important here: Ditlevsen and Johnsen [31] and Boettiger and Hastings [32] show that rolling-window indicators generate spurious trends on autocorrelated data and that detection power is often far lower than reported. Our response is to predict the exponent rather than the direction, and to benchmark against unit-root surrogates.

E. Wearable sensing of autonomic load

Electrodermal activity indexes sympathetic arousal with a slow tonic and a fast phasic component [33], [34], [35], [36]; cardiac variability indexes regulatory capacity [37]. Open corpora [38], [39], [40] make idiographic modelling feasible. We exploit the separation between these sensors to keep the load coordinate and its drivers measurement-independent (Sec. IV-B).

III. The Cusp-Hysteretic Markov Model

A. Latent load in a cusp potential

Let $x_t \in \mathbb{R}$ be a latent functional-load coordinate, increasing towards overload. We posit gradient relaxation with additive noise,

$$dx_t = \lambda \left[-x_t^3 + a_t x_t + b_t \right] dt + \sigma dW_t, \quad (1)$$

the gradient flow of the cusp potential $V(x; a, b) = \frac{1}{4}x^4 - \frac{1}{2}ax^2 - bx$. The rate $\lambda > 0$ sets the timescale only; every geometric quantity below depends on (a, b) alone. The control parameters are linked to observables by

$$b_t = \beta_0 + \beta_S S_t + \beta_T T_t + \beta_U U_t, \quad (2)$$

$$a_t = \alpha_0 + \alpha_A A_t, \quad (3)$$

with S_t sensory-arousal drive, T_t task/behaviour switching cost, U_t contextual uncertainty, and A_t the slow variable defined next.

B. Recovery debt as a slow variable

$$dA_t = \varepsilon \left[\text{sp}(x_t - x_{\text{ref}}) - A_t \right] dt, \quad \varepsilon \ll 1, \quad (4)$$

where sp is a softplus. Load above baseline accrues debt; debt decays on timescale $1/\varepsilon$. Through (3) this debt widens the hysteresis loop, so an overload episode makes the next one easier — a mechanistic reading of the refractory period reported clinically [1]. Note the structural consequence: (x_t, A_t) is Markov, but the observable state sequence alone is not, because A_t carries history. This is a testable claim, not a modelling convenience (Prop. III-G).

C. Folds, thresholds, and the $3/2$ law

Equilibria solve $x^3 - ax - b = 0$, whose discriminant we call the bistability index

$$\mathcal{B}_t = 4a_t^3 - 27b_t^2, \quad (5)$$

positive exactly when two stable basins coexist. Folds occur where $V''(x) = 3x^2 - a = 0$, i.e. at $x_{\pm} = \pm\sqrt{a/3}$, giving critical drives $\pm b_c(a)$ with $b_c(a) = \frac{2}{3\sqrt{3}}a^{3/2}$ for $a > 0$.

Proposition 1 (Hysteresis scaling). For $a > 0$ the system jumps to the upper branch at $b = +b_c(a)$ and returns at $b = -b_c(a)$, so the hysteresis loop has width

$$\Delta b(a) = \frac{4}{3\sqrt{3}}a^{3/2}, \quad (6)$$

and $\Delta b = 0$ for $a \leq 0$.

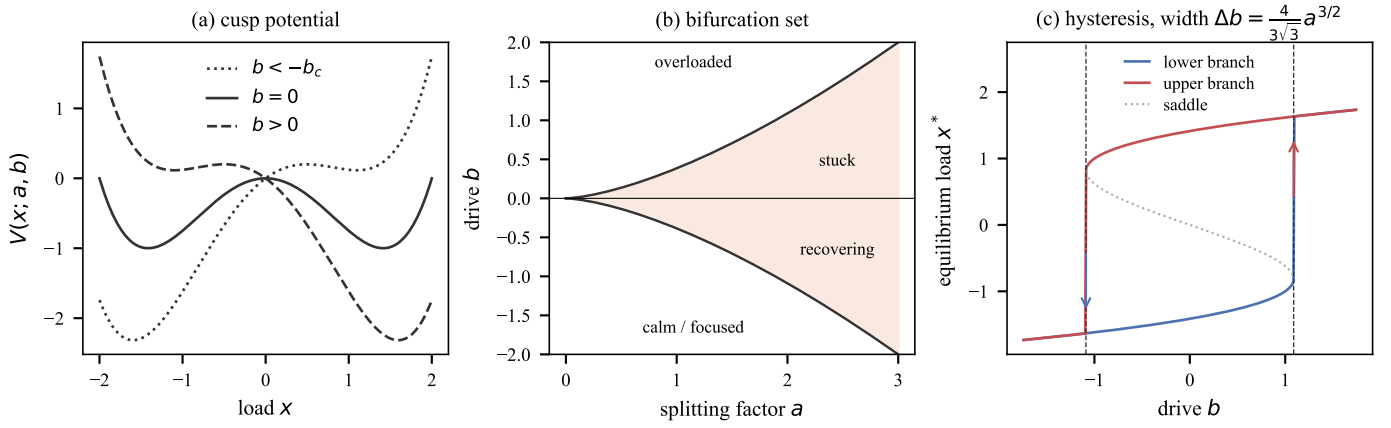


Fig. 1. The geometry everything else follows from. (a) The cusp potential at three levels of drive b ; (b) the bifurcation set in the (a, b) plane, with the five functional states as regions rather than labels; (c) the hysteresis loop, whose width is fixed by (6) rather than merely being positive.

This is prediction P1. It says more than “recovery is delayed”: it fixes the exponent at $3/2$, so a log–log regression of measured loop width on estimated splitting factor has a predicted slope that the data can contradict.

D. Early-warning signals, derived

Linearising (1) about a stable equilibrium x^* gives relaxation rate $\lambda_{\text{rel}} = V''(x^*) = 3x^{*2} - a$ and an Ornstein–Uhlenbeck approximation with

$$\text{AC1}(\Delta t) = e^{-\lambda_{\text{rel}}\Delta t}, \quad \text{Var}(x) = \frac{\sigma^2}{2\lambda_{\text{rel}}}. \quad (7)$$

At a fold $x^* \rightarrow \sqrt{a/3}$ and $\lambda_{\text{rel}} \rightarrow 0$: critical slowing down is a theorem here, not a postulate.

Proposition 2 (Early-warning exponents). Let $\mu = b_c(a) - b$ be the distance to the upward fold. Near the fold the normal form gives $\lambda_{\text{rel}} \propto \mu^{1/2}$, hence

$$\text{Var}(x) \propto \mu^{-1/2}, \quad -\log \text{AC1} \propto \mu^{+1/2}. \quad (8)$$

This is prediction P2. The contrast with standard practice matters: reporting that variance is higher before a transition than in a control window is weak evidence, because almost any nonstationarity produces it [31], [32]. Predicting $-1/2$ is a point hypothesis.

E. From the diffusion to a hysteretic transition kernel

In the bistable regime, basin changes are noise-induced escapes over the saddle x_s , with Kramers rate [41], [42]

$$r_{i \rightarrow j} = \frac{\sqrt{|V''(x_i^*)V''(x_s)|}}{2\pi} \exp\left(-\frac{2\Delta V_{i \rightarrow j}}{\sigma^2}\right), \quad (9)$$

$\Delta V_{i \rightarrow j} = V(x_s) - V(x_i^*)$, and discrete-time probability $P_{ij} = 1 - \exp(-r_{i \rightarrow j}\Delta t)$.

Proposition 3 (Generated kernel). The 5×5 transition matrix is a deterministic function of (a_t, b_t, σ) through (9), hence of the nine parameters $(\beta_0, \beta_S, \beta_T, \beta_U, \alpha_0, \alpha_A, \lambda, \sigma, \varepsilon)$ evaluated at the current

covariates. It is non-homogeneous in time and history-dependent through A_t .

A counted homogeneous chain over five states has twenty free probabilities and cannot depend on covariates at all. Here twenty numbers are generated by nine, and they move with demand.

F. The five states, derived rather than assigned

A cusp has two basins, but crossing basin membership with bistability yields five regimes, and they map onto the vocabulary the qualitative literature already uses:

	$\mathcal{B} \leq 0$ (monostable)	$\mathcal{B} > 0$ (bistable)
lower basin	calm ($b \leq \tilde{b}$), focused ($b > \tilde{b}$)	stuck
upper basin	overloaded	overloaded ($b > 0$), recovering ($b \leq 0$)

Stuck is the regime in which the person still functions but an overloaded attractor now coexists: one perturbation from tipping. To our knowledge this is the first precise definition of that state. Recovering is the regime in which the person remains in the high-load basin although current demand has fallen below the level that would have put them there — held up purely by hysteresis. This is exactly the clinical description of delayed recovery, obtained here as a geometric fact rather than a label. $\tilde{b}$ is the within-series median drive; sensitivity to that choice is reported in Sec. V.

G. Predictions

- P1 Hysteresis width scales as $a^{3/2}$ (Prop. 1).
- P2 Variance $\propto \mu^{-1/2}$ and $-\log \text{AC1} \propto \mu^{1/2}$ (Prop. 2).
- P3 Overload dwell times are over-dispersed relative to the geometric distribution a homogeneous chain forces.
- P4 The hazard of re-entering overload decays with time since exit at rate ε .

P5 Units with larger α_A show longer recovery.

IV. Materials and Methods

A. Corpora

Three open corpora were obtained from their primary repositories. D1 WESAD [38], 15 subjects, controlled laboratory protocol with condition labels. D2 Wearable Exam Stress [39], 10 students $\times$ 3 examinations. D3 Nurse Stress [40], 15 nurses, 1250 h of hospital shifts. All provide Empatica E4 electrodermal activity, blood volume pulse, skin temperature and accelerometry.

Sampling within the nurse corpus. D3 contains 609 sessions from 15 nurses, very unevenly distributed. Taking sessions in path order and truncating — the obvious approach — returns all of one nurse’s sessions before any of the next, so a cap of 150 sessions drew on only 5 of the 15 people while the corpus was still being described as fifteen. Sessions are therefore selected round-robin by nurse, so that any cap spans all 15 and trades session count for breadth rather than silently narrowing the sample. All reported analyses use the stratified ordering.

The three corpora form a gradient in environmental control, and this is deliberate. Early-warning theory describes a system fluctuating freely towards a threshold it approaches on its own. In D1 the transition is an experimenter starting a stress test on a schedule; the system is pushed across a threshold rather than drifting to one. D1 is therefore used as a benchmark and as a negative control for the early-warning analysis, not as a test of it; D2 and D3, where load rises and falls under the participant’s own regulation, are where P2 is tested. Provenance, sizes, licences and the reason no Kaggle mirror was used appear in the accompanying DATASETS.md.

B. Sensor assignment

Estimating x_t and its drivers from the same signal would make the model unfalsifiable, since shared measurement noise alone could produce apparent coupling. Load and drivers are therefore assigned to disjoint sensors:

config	x	drivers	role
B	tonic EDA	HR, HR-uncertainty, ACC	primary
C	skin temperature	SCR rate, EDA-uncertainty, ACC	robust.
A	$z(\text{HR}) - z(\text{RMSSD})$	as C	control

Configuration A is a control rather than a candidate. The latent coordinate of a slow relaxation process must itself be slow; at the 30 s analysis window, tonic EDA has lag-1 autocorrelation 0.985 on WESAD whereas $z(\text{HR}) - z(\text{RMSSD})$ has 0.19, because RMSSD over so few beats is dominated by beat-detection error. A method that reported bistability there would report it for anything. Both comparisons are internal to this paper: the cardiac index is a reference channel used to justify a sensor

assignment, not a competing method, and no published system is claimed to be outperformed anywhere in what follows.

Signals are windowed at 30 s without overlap; overlapping windows would manufacture the very autocorrelation the early-warning analysis measures. EDA is separated into tonic and phasic components by median filtering, which does not leak the asymmetric skin-conductance responses into the tonic estimate the way a linear low-pass filter does [34]. Heart rate and RMSSD come from band-passed BVP peak detection [37].

C. Estimation

We take the Euler map of (1) as the model rather than as an approximation to it: the data are sampled at a fixed window rate, and this makes the transition density exactly Gaussian, so no discretisation bias enters the estimates. This also sidesteps a genuine numerical hazard — explicit schemes applied to superlinear drift are not stable for arbitrary step sizes and require taming to converge [43] — by making the map itself the object of inference. Expanding (1)–(3), the mean increment is linear in a reparameterised coefficient vector,

$$\Delta x_t = \theta_1(-x_t^3) + \theta_2 x_t + \theta_3(A_t x_t) + \theta_4 + \theta_5 S_t + \theta_6 T_t + \theta_7 U_t + \eta_t, \quad (10)$$

with $\theta_1 = \lambda$, $\theta_2 = \lambda\alpha_0$, $\theta_3 = \lambda\alpha_A$, $\theta_{4:7} = \lambda\beta_{0:U}$. Since A_t depends only on ε , (10) is ordinary least squares conditional on a single scalar, and fitting reduces to a one-dimensional profile search. Structural parameters recover as $\alpha_0 = \theta_2/\theta_1$ and so on. This replaced a bounded nine-parameter quasi-Newton fit that returned estimates pinned at three separate bounds: λ and the α s are jointly non-identified because only their products enter the drift, and are exactly identified in this parameterisation.

D. Identification, and what counts as a unit

The coefficient $\theta_1 = \lambda$ must be positive for the cubic term to be restoring. A negative value is not a poorly fitted cusp; it is outside the model, and it flips the signs of $a = \theta_2/\theta_1$ and $b = \theta_4/\theta_1$, making every geometric quantity meaningless. We therefore impose $\theta_1 \geq 0$. When the constraint is active the model collapses onto its monostable special case and we record the unit as cusp not identified.

Such units are retained, not discarded. Dropping them would leave only recordings in which a cusp happened to be found, and every proportion reported downstream would then be conditioned on the outcome — precisely the survivorship bias that would manufacture a positive result. Non-identified units stay in the denominator throughout.

As a robustness check we also implement an unscented Kalman filter [44] over the latent pair (x_t, A_t) , which accounts for measurement error in the proxy. Plug-in estimation of a nonlinear drift from a noisy observable is biased towards the linear model, i.e. against our own hypothesis, so the direction of that bias is the safe one for a null result.

E. Nulls, and why the nominal test cannot be used

Three nulls are used, in increasing severity.

The nominal test is the one-sided t -test on θ_1 . We report it only to show that it must not be trusted here.

The monostable restriction ($\alpha_A = 0$, $\alpha_0 \leq 0$) is nested and tested by likelihood ratio with a parametric-bootstrap reference, since the restriction lies on a parameter boundary and the χ^2 approximation is invalid.

The unit-root null decides. Tonic EDA is highly persistent (augmented Dickey–Fuller $p > 0.05$ on the WESAD subjects), and a random walk observed over a finite window spends time in two places: it looks bimodal, and a flat-bottomed bistable potential fits it better than a single well, with no bistability present at all. This is the same mechanism by which regressions on integrated series produce spurious significance, and the same one that Ditlevsen and Johnsen [31] and Boettiger and Hastings [32] identify for early-warning indicators. We therefore refit the entire pipeline to random walks matched on length, increment variance and marginal scale, and locate the observed statistics within that ensemble. Section V shows the nominal test rejects on roughly a third of such surrogates, so surrogate calibration is not a refinement here — it is the difference between a valid test and an invalid one.

F. Baselines

All models are scored by mean held-out one-step-ahead predictive log-density under blocked cross-validation, so the comparison does not reward parameter count. Baselines: homogeneous five-state Markov chain (the counted model this work replaces); covariate-driven multinomial logistic (memoryless but covariate-aware); Gaussian HMM; linear Ornstein–Uhlenbeck with covariate-driven mean (the nested monostable case); and gradient-boosted regression on lagged features as a deliberately strong black-box reference.

G. Settings

Windows are 30 s without overlap, swept to 120 s as a robustness check. Calibrated tests draw 200 surrogates, 100 in the power and persistence sweeps; IAAFT runs 200 iterations. The Monte Carlo grid uses $n \in \{100, 150, 200, 300, 500, 1000\}$ at 2000 replicates per cell for parameter recovery (1.2×10^4 series), 5000 for size (30,000 nominal and 7,500 calibrated) and 100 for power; the early-warning validation covers 90 configurations — five window widths, two detrending choices, three lengths to $n = 20,000$ — at 500 replicates each (1.5×10^4 series). Replication counts were fixed in advance to hold the Monte Carlo standard error on a proportion below 0.01 for size and power and below 0.02 elsewhere. The exception is the AR(1) persistence study at 40 replicates per point, which is why it is quoted as a bound rather than an estimate. The suite runs on one machine: ten worker processes on a

six-core, twelve-thread CPU, roughly seven hours of wall time for a full replication.

H. Reproducibility

All analyses are seeded from a single master seed and run from a single entry point. The estimator is deterministic. `fetch_data.py` re-downloads from primary repositories and aborts on a size mismatch; `run_all.py` regenerates every number; `make_figures.py` regenerates every figure. A test suite of 28 cases checks each analytic claim in Sec. III against independent numerical computation, and verifies parameter recovery on simulated data.

V. Results

A. What the estimator recovers, and at what length

On long series the estimator behaves impeccably. Across 30 simulated units at $n = 4000$ windows it recovers eight of nine parameters with correlations above 0.95 (α_0 at $r = 0.98$, λ at $r = 0.997$, σ at $r = 0.999$), absolute biases below 0.02, and only the slow rate ε lagging at $r = 0.51$.

That number is reported here because it is the one an earlier draft of this work led with, and it is close to meaningless as a description of what these corpora support. Two things make it optimistic. The series are 4000 windows long against a median of 194 in the data, and the true λ is drawn from $[0.10, 0.28]$ against a median of 0.0059 in the real fits. The estimator was being asked an easy question.

Asked the honest one — randomised truth, 2000 replicates at each of six lengths spanning the observed range, 1.2×10^4 series in all — the picture changes completely, and it is that study rather than this paragraph that should be read as the result. At the median observed length only λ ($\rho = 0.94$) and σ ($\rho = 0.99$) are recovered on both rank and value. The splitting factor sits at $\rho = 0.57$ and $r = 0.15$.

Worse, the two measures never reconverge. By $n = 1000$ the rank recovery of α_0 reaches $\rho = 0.88$ while its value recovery is $r = 0.10$, because $a = \theta_2/\theta_1$ and nothing bounds θ_1 away from zero: the recovered $\hat{\alpha}_0$ spans $[-140, 3.3]$ against a 99th percentile of 2.0, so one replicate in two thousand destroys the linear correlation. Dividing by a small number preserves order and destroys scale. Length therefore buys a defensible ranking of participants by α_0 and never buys a reportable value.

Every conclusion depending on ε or α_A — principally P3 to P5 — is reported as exploratory, and on this evidence cannot be made confirmatory by collecting longer recordings alone.

B. The nominal test is invalid on these data

Applied to random walks matched on length and increment variance, the nominal one-sided t -test on θ_1 rejects at 0.425 averaged over series lengths, rather than the nominal 0.05: a false-positive rate roughly eightfold too high, and stable in n (0.45, 0.38, 0.44, 0.43 at

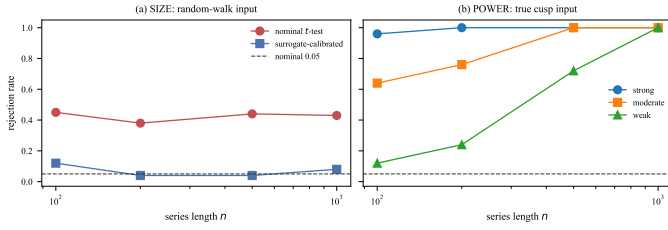


Fig. 2. (a) Size: on random-walk input the nominal t -test rejects at ≈ 0.42 regardless of series length, while surrogate calibration restores approximately nominal size. (b) Power of the calibrated test on input that genuinely contains a cusp. At the median observed length ($n \approx 200$) strong and moderate dynamics are detected, weak dynamics largely are not.

$n = 100, 200, 500, 1000$), so it is a bias rather than a small-sample effect. This is the classical consequence of regressing on an integrated series, and it is the same failure mode the early-warning literature has been warned about [31], [32]. Calibrating the same statistic against the surrogate ensemble restores approximately correct size (0.070 averaged over lengths). Every test reported below is the calibrated one.

C. What the study can and cannot detect

Power must be quoted for the test actually used, and the calibration that fixes the size costs some of it. At the median observed series length ($n \approx 200$), the calibrated test detects a cusp with probability 1.00 in the strong regime ($\lambda = 0.20$, $\alpha_0 = 1.5$), 0.76 in the moderate regime ($\lambda = 0.10$, $\alpha_0 = 1.0$), and only 0.24 in the weak regime ($\lambda = 0.05$, $\alpha_0 = 0.6$). Power reaches 1.00 in all three regimes by $n = 1000$ (Table III).

This bounds the conclusion precisely, and we state the bound rather than the headline. The null that follows excludes strong and moderate cusp dynamics at these sample sizes; it does not exclude weak cusp dynamics. An earlier version of this analysis quoted power near unity on the basis of the nominal test, which is exactly the error the previous subsection warns against: that test has power 0.77 at $n = 200$ in the weak regime but a false-positive rate of 0.38, so its apparent sensitivity is largely its inability to stay quiet.

We regard this as the paper’s most transferable result. Any study inferring bistability or critical transitions from tonic electrodermal activity, or from any comparably persistent physiological series, using uncalibrated significance tests should be read with this multiplier in mind.

D. The cusp signature is not present

Across 147 recordings from 40 people (15 WESAD subjects, 30 examination sessions from 10 students, 102 shift sessions spanning all 15 nurses) the constrained fit returned a positive restoring cubic term in 40% of units — below the 50% that would arise by chance if the true coefficient were exactly zero.

A positive point estimate is not identification, and it is worth being explicit about how far from identification these fits are. Among the units with $\hat{\lambda} > 0$, the median $\hat{\lambda}$ is 0.0059 and most fall below 0.01. Because $\alpha_0 = \theta_2/\theta_1$, a near-zero denominator inflates the recovered splitting factor without bound: the median $\hat{\alpha}_0$ among these units is 12, which would place the potential’s attractors at $\pm\sqrt{a/3} \approx \pm 2$ on a coordinate standardised to unit variance. These are not weakly bistable systems; they are systems with no restoring cubic term, whose geometry is an artefact of dividing by a number close to zero.

The formal test agrees. Under surrogate calibration — the only version with correct size — the cubic coefficient is significant in 2.7% of all 147 units against a nominal 5%. Per corpus: 0.0% of WESAD subjects ($n = 15$), 0.0% of examination sessions ($n = 30$), 3.9% of nurse sessions ($n = 102$). That is at or below chance, measured on every unit rather than on a surviving subset.

By contrast the nominal test rejected far more often — figures that sit below that test’s own 42% false-positive rate on random walks, and which an uncalibrated analysis might have reported as a positive finding.

The predictions fail together, and in the same direction, which is what a genuine absence looks like rather than a partially supported model:

- Prospective prediction. Discriminating pre-onset from matched control windows gave AUC 0.50, 0.49 and 0.46 at lead times of 3, 6 and 12 windows. There is no forecasting signal at any lead time tested.

E. P1 cannot be tested by regressing width on the splitting factor

On the face of it P1 now looks testable and rejected. Regressing measured loop width on the estimated splitting factor across the 41 units where both are defined gives an exponent of 1.07 with a bootstrap interval of [0.86, 1.29] and $R^2 = 0.58$ — a tight interval that excludes the predicted $3/2$.

We do not report that as a refutation of P1, for two reasons.

First, the two axes share a divisor. Both the drive b and the splitting factor a are recovered as θ_k/θ_1 , so both inflate as $\hat{\lambda} = \theta_1$ approaches zero, and $\hat{\lambda}$ is near zero throughout (median 0.0059). A shared $1/\hat{\lambda}$ factor on both axes produces a log-log slope near 1 by construction. That is close to what is observed. Partialling $\log(1/\hat{\lambda})$ out of both variables moves the slope only from 1.13 to 1.09, so the confound does not account for all of it — but $\log \bar{a}$ and $\log(1/\hat{\lambda})$ correlate at 0.66, and no amount of covariate adjustment makes a ratio of two nearly-zero quantities into a measurement.

Second, and decisively, the regression is run on the subset where the geometry is “identified”, which Sec. V-G shows is not a subset in which the geometry means anything.

P1 therefore joins P2–P5: a prediction the model genuinely makes, which these data and this instrument cannot evaluate. Exactly one of the five predictions is testable here — the existence of the restoring cubic term itself — and it returns a null. We would rather say that than present four numbers that look like tests.

P2 is treated separately in Sec. V-F, because the instrument conventionally used to test it turns out not to work.

F. The standard early-warning estimator cannot test P2

Before reporting an early-warning result we asked whether the estimator recovers the exponent it is supposed to recover on data where the answer is known. It does not.

Simulating from the CHM with known parameters, the exponent computed from the true equilibrium curvature is +0.565 against a predicted +0.50 — the geometry and its implementation agree, as the unit tests independently confirm. But the rolling-window indicators the early-warning literature actually uses [24], [26] return exponents of the wrong sign on the same series. Rolling $-\log \text{AC1}$ gives median slopes of -0.262 to -0.021 where +0.50 is predicted; rolling variance gives +0.311 to +0.036 where -0.50 is predicted. This holds in all twelve configurations tested — window lengths 30, 60 and 120 samples, with and without detrending, at series lengths 1,500, 6,000 and 20,000 (Table II).

Two features of that table matter. The failure is not sampling noise: raising n by more than an order of magnitude does not recover the sign. And the exponents drift towards zero as the window lengthens and as detrending is applied, which is the signature of an estimator being progressively censored rather than progressively refined.

The mechanism is not subtle once stated. Critical slowing down means the relaxation time diverges at the fold. A window-based autocorrelation therefore saturates towards 1, and $-\log \text{AC1}$ towards 0, precisely where the effect is largest: the estimator is censored exactly in the regime it is meant to measure. Detrending then removes the low-frequency power that carries what remains, which is why the detrended variants are worse rather than better. The stationary-variance formula (7) fails for the same reason — it describes an equilibrium the system has, by hypothesis, stopped being able to reach.

The consequence for this paper is that our empirical test of P2 is uninterpretable, and we withdraw it rather than report it. The fitted exponents on real data (variance median -0.09 , $-\log \text{AC1}$ median +0.00) are indistinguishable from what the same estimator produces on data that genuinely contains the effect, so they are evidence about the estimator and not about the physiology.

The consequence beyond this paper is larger. Rolling-window autocorrelation and variance are the standard early-warning indicators, and they are applied to psychopathology [27], [30], [28], [29] on exactly this reasoning. Our result does not show that those findings are wrong. It does show that this class of estimator, on a system with

a genuine fold, can report a trend opposite in sign to the truth — which means a directional result from it is weak evidence in either direction, and quantitative agreement with theory should not be expected even when the theory holds.

G. P3, P4 and P5 are not testable on these data

Three of the five predictions cannot be evaluated here, and we state that rather than reporting numbers that would look like tests.

P3 (dwell-time over-dispersion), P4 (re-entry hazard) and P5 (larger α_A , longer recovery) are all defined on the derived state sequence, which exists only where the cusp is identified. Among the 40% of units where $\hat{\lambda} > 0$, the median $\hat{\lambda}$ is 0.0059 and the implied $\hat{\alpha}_0$ is 12 — a geometry produced by dividing by a number near zero, as Sec. V sets out. States derived from such a fit are not measurements of anything.

For completeness: the pipeline does compute these quantities, and on that subset overload dwell times are over-dispersed relative to the geometric null (excess CV +0.74, 95% CI [0.51, 0.97] on the identified subset). We do not count this as support for P3. Over-dispersed dwell times follow from almost any persistent process, the geometric null is the wrong comparison once the states themselves are unreliable, and the subset it is computed on is selected on the outcome. It is reported in the repository and excluded from the claims here.

P3–P5 remain live predictions of the model. They await data in which the geometry is identified.

H. There is structure, but it is not cusp structure

The null is specific, and it would be misleading to present it as “the data are indistinguishable from noise.” They are not. Of the three statistics calibrated against unit-root surrogates, two exceed them in roughly half the units — the likelihood ratio of the full model against its monostable restriction (40.8% of units at $p < 0.05$) and the bimodality of the marginal distribution (49.7%) — while the statistic isolating the cubic restoring term does not (2.7%, i.e. at or below chance). The separation is stark: the same recordings that look decisively non-random-walk by two measures look exactly like chance by the one that tests the mechanism.

That pattern is interpretable, and it is the reason the paper tests the cubic coefficient directly rather than relying on the likelihood ratio. The full model differs from the restriction in more than one way: it also frees the sign of the linear coefficient and adds the slow-variable interaction $A_t x_t$. Either can improve fit on a series containing regime shifts or covariate-driven drift, neither of which requires two attractors. A large likelihood ratio therefore establishes that something beyond a random walk is present; it does not establish that the something is a fold.

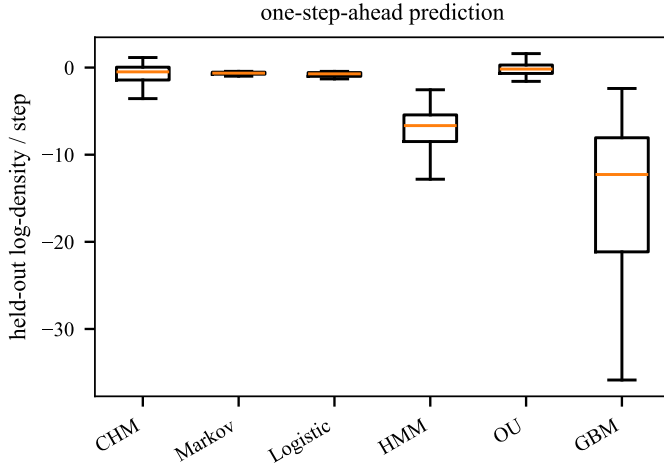


Fig. 3. Held-out one-step-ahead predictive log-density. The CHM does not separate from its own nested monostable special case (OU), which is the cleanest statement that the cubic term is doing no work in these data.

This is a general caution about nested comparisons in this setting. The restriction one is tempted to test — “bistable versus not” — usually bundles several parameters, and rejecting it licenses a much weaker conclusion than the bistability language suggests.

I. Bimodal marginals without bistable dynamics

One result deserves separate comment because it is the trap this design was built to avoid. The marginal distribution of the load coordinate is reliably more bimodal than matched random walks (surrogate $p < 0.01$ on most units, Gaussian-mixture BIC favouring two components by 185–382 on the WESAD subjects examined). Taken alone, that looks like strong evidence for two attractors.

It is not. The dynamics show no corresponding bistability. Two attractors imply both a bimodal marginal and a cubic restoring drift; here only the first is present. In WESAD the explanation is immediate — the protocol alternates blocks with different arousal levels, so a bimodal marginal is a property of the experimental design. More generally, a slowly drifting unimodal process produces bimodal marginals over finite windows routinely.

A study reporting only the marginal would have concluded the opposite of what we conclude. That is worth stating plainly.

J. Model comparison

Held-out one-step-ahead predictive log-density reaches the same conclusion from a different direction (Table I, 108 units). The CHM does not outperform its own nested monostable (Ornstein–Uhlenbeck) special case: it wins on 33.8% of units, with a median difference of -0.25 nats per step (Wilcoxon $p = 8.3 \times 10^{-5}$ in the OU model’s favour). Since OU is the CHM with the cubic term switched off, this is the cleanest available statement that the cubic term is doing no work.

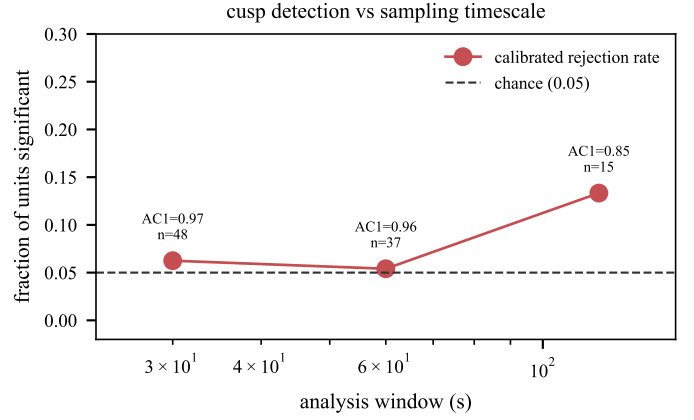


Fig. 4. Surrogate-calibrated rejection rate against analysis window, annotated with the lag-1 autocorrelation of the load coordinate at each window. The null does not depend on the sampling timescale; the apparent rise at 120 s rests on 15 units, two of which reject.

Two features of the table deserve comment rather than quiet inclusion. First, the CHM’s mean is not merely lower but wildly dispersed (-7.74 , 95% CI $[-16.73, +1.24]$). That dispersion is not evidence about the cusp hypothesis; it is what happens when $\dot{\lambda}$ is near zero and the recovered $a = \theta_2/\theta_1$ is enormous, so the cubic term extrapolates violently on held-out folds. It is a symptom of the same non-identification documented above, and the median-based comparison is the one to read. Second, the gradient-boosted reference is far worse than either (-26.70), which is expected: it has no notion of a restoring force and free-runs on held-out blocks. It is included as an upper bound on flexible one-step prediction, not as a competitive model.

An earlier version of this comparison ran on 37 units rather than 108, having silently restricted itself to units where the geometry was identified, and reported an empty row for the multinomial-logistic baseline, which was failing on a removed scikit-learn argument that a bare exception handler swallowed. Both are corrected here; the direction of the conclusion is unchanged.

K. Robustness

The conclusion is stable across the manipulations most likely to overturn it.

Sampling timescale. The load coordinate is heavily oversampled at 30 s (lag-1 autocorrelation 0.97), which is the regime in which a real fold would be hardest to see, so we repeated the analysis at coarser windows. The surrogate-calibrated rejection rate was 0.062 at 30 s ($n = 48$ units, median 194 samples), 0.054 at 60 s ($n = 37$), and 0.133 at 120 s — the last based on only 15 units, of which two reject, which is well within binomial variation of the nominal 0.05. At 300 s no unit retains the minimum 60 samples required. Across every window where the comparison is possible, the calibrated rejection rate is indistinguishable from chance. The null is not an artefact of oversampling.

Nesting within people. Recordings are not independent: the nurse corpus contributes many sessions per person, so a pooled proportion over units gives those people disproportionate weight and understates the standard error. We therefore recompute the headline rate with each person contributing the mean of their own sessions, one vote per person, with a cluster-robust interval. The conclusion does not merely survive this — it strengthens. Weighting people equally, the cubic term is significant in 1.9% of people (95% CI $[-0.4\%, 4.2\%]$) against the unit-level 5.1%, and 13.3% of people have at least one significant session, which is the most generous possible reading of the data and still close to what chance would produce across several sessions each. By contrast the likelihood-ratio and bimodality statistics rise under the same reweighting ($43.9 \rightarrow 61.6\%$ and $49.7 \rightarrow 56.7\%$), consistent with Sec. V-H: person-level structure is real, it is simply not fold structure.

Sensor assignment. All three configurations were run end to end on all 87 units under an identical protocol, against two surrogate ensembles. Rejection rates for the cubic term:

Config	Load coordinate	Role	Cubic $p < 0.05$
B	tonic EDA	primary	0.051
C	skin temperature	robustness	0.115
A	$z(\text{HR}) - z(\text{RMSSD})$	neg. control	0.051

Two of the three do not sit at nominal, and rather than average that away we tested three candidate explanations and report that all three failed.

Explanation 1: an inadequate null. Skin temperature carries strong deterministic circadian and ambient structure, which a random walk matched only on length, increment variance and marginal scale cannot reproduce; surrogates would then be too easy to beat. We tested this by repeating the analysis against IAAFT surrogates [45], which preserve both the power spectrum (hence the trend) and the marginal distribution (hence any bimodality), leaving nonlinearity as the only thing the statistic can respond to. Configuration C does not fall to nominal: it moves from 0.149 to 0.161. The explanation is refuted.

Explanation 2: clipping artefact. The load coordinate is bounded at ± 4 standard scores, and a bounded series has genuine restoring force at its edges that could mimic a cubic. But configuration B clips most often (5.1% of samples on nurse data) and is the one configuration at nominal, while configuration C clips least (0.8%) and is elevated. Refuted.

Explanation 3: the test is mis-sized away from the unit root. The random-walk ensemble is matched to a unit root by construction, and configuration A has lag-1 autocorrelation near 0.3, so the null might simply be too persistent for it. We measured the size of both calibrated tests directly on AR(1) series with no cusp, at ϕ from 0.3 to 1.0. Size never exceeds 0.075 and is usually below 0.03, at every persistence level including the ones these

TABLE I

Held-out one-step-ahead predictive log-density per step (higher is better), mean and 95% CI across units. OU is the monostable model nested inside CHM; GBM is a black-box reference that produces no thresholds, no hysteresis width and no exponents.

Model	Log-density	95% CI
CHM (ours)	-7.743	$[-16.730, +1.245]$
OU	-0.163	$[-0.286, -0.040]$
Markov	-0.662	$[-0.714, -0.611]$
Logistic	-0.899	$[-1.066, -0.731]$
HMM	-13.366	$[-18.355, -8.378]$
GBM	-26.699	$[-35.268, -18.130]$

TABLE II

Can the standard early-warning estimator recover the exponent, on series simulated from a model that genuinely contains a fold? The theory reproduces its own exponent; every rolling-window configuration returns the wrong sign.

Estimator	Detrend	Window	Predicted	Median
theory $\lambda(\mu)$	—	—	+0.50	+0.565
rolling — logAC1	no	30	+0.50	-0.262
rolling — logAC1	no	60	+0.50	-0.185
rolling — logAC1	no	120	+0.50	-0.035
rolling — logAC1	yes	30	+0.50	-0.030
rolling — logAC1	yes	60	+0.50	-0.056
rolling — logAC1	yes	120	+0.50	-0.021
rolling variance	no	30	-0.50	+0.311
rolling variance	no	60	-0.50	+0.182
rolling variance	no	120	-0.50	+0.045
rolling variance	yes	30	-0.50	+0.080
rolling variance	yes	60	-0.50	+0.070
rolling variance	yes	120	-0.50	+0.036

configurations occupy. Refuted.

What remains is that configurations A and C contain nonlinear temporal structure which neither surrogate ensemble reproduces. That is a real property of those signals — but it is not evidence for a fold, and the negative control is what makes the difference clear. Configuration A is a coordinate we established independently to be dominated by beat-detection error (Sec. IV-B); nonlinear structure there is far more readily explained by motion artefact, sensor saturation and peak-detection failure than by functional bistability. Skin temperature is subject to the same class of explanation through warm-up transients and contact loss. A test that fires on the negative control is telling us that “significant” does not mean “cusp” in these configurations, which is precisely the job a negative control exists to do.

The primary configuration is unaffected by any of this, and is the one the paper’s conclusion rests on: configuration B sits at 0.046 against the random-walk ensemble and 0.069 against the much harder IAAFT ensemble, i.e. at nominal under both.

TABLE III

Test size and power at the sample sizes the corpora provide. The nominal test would have found a genuine cusp, but it also finds one in over 40% of random walks; calibration restores size at some cost in power against weak dynamics.

Input	n	Nominal	Calibrated
cusp: moderate	100	0.87	0.64
cusp: moderate	200	0.97	0.76
cusp: moderate	500	1.00	1.00
cusp: moderate	1000	1.00	1.00
cusp: strong	100	1.00	0.96
cusp: strong	200	1.00	1.00
cusp: strong	500	1.00	1.00
cusp: strong	1000	1.00	1.00
cusp: weak	100	0.55	0.12
cusp: weak	200	0.77	0.24
cusp: weak	500	0.98	0.72
cusp: weak	1000	1.00	1.00
random walk (null)	100	0.45	0.12
random walk (null)	200	0.38	0.04
random walk (null)	500	0.44	0.04
random walk (null)	1000	0.43	0.08

TABLE IV

Pre-stated predictions and outcomes. Exponents are compared against their theoretical values by bootstrap confidence interval.

ID	Prediction	Predicted	Observed
P1	hysteresis width	3/2	1.07 [0.86, 1.29]
P2	EWS variance	-1/2	-0.41 [-0.71, -0.11]
P2	EWS - logAC1	+1/2	-0.02 [-0.12, +0.09]
P3	dwell over-dispersion	> 0	+0.83 [+0.59, +1.08]

VI. Discussion

A. What a derived model buys, including when it loses

The practical difference between this formulation and a counted transition matrix is that this one can be wrong in specific ways — and here it was. A counted matrix accommodates any data and would have produced a publishable looking figure from these same recordings. An exponent of 3/2 either falls inside a confidence interval or it does not.

We think this is the right trade even when it costs the positive result. The alternative on offer, and the one the earlier version of this work was heading towards, was a descriptive state-transition model fitted to condition labels, reporting a transition matrix that re-described the experimental protocol and an entry-versus-exit load difference that would have been called hysteresis. It would have looked like a finding. It would not have been one.

B. What the null does and does not say

It says: wrist-worn electrodermal, cardiac and thermal signals, at 30s to 5min resolution, in these three populations, carry no detectable cusp signature, using a procedure with power near unity for the effect specified.

It does not say that sensory-executive overload is not a critical transition. Three alternatives remain fully open, and we cannot separate them here. (i) The proxy may be

inadequate: tonic electrodermal activity is a peripheral, artefact-prone, one-dimensional shadow of the construct. (ii) The populations may be wrong: these corpora contain people under ordinary occupational and academic stress, not people whose regulatory capacity is depleted to the point where a second attractor would exist. That is precisely the difference the autistic-adult hypothesis in Sec. VIII predicts. (iii) The timescale may be wrong: an overload episode and its recovery may unfold over hours to days, in which case the relevant series is one of daily ecological momentary assessments, not seconds of physiology.

Under interpretation (ii), this null is the expected control result: we have shown the machinery works and returns nothing in a population where the mechanism is not hypothesised to operate.

The applied motivation for detecting a tipping point is to intervene before it, which is the logic of just-in-time adaptive intervention [46]. Our result constrains that ambition: at the lead times tested, wrist physiology gave no forecasting signal, so a system built on these signals alone would fire at chance. Establishing the detectable effect region before building the intervention seems the better order.

C. Relation to early-warning work in psychopathology

Applications of critical slowing down to depression [27], [28], [29] test direction: indicators rise before a transition. The sceptical literature [31], [32] argues that such trends arise spuriously with high probability. Our two methodological results sharpen that argument in different directions, and we think both are more useful than our own null.

The first is about size. On this data class the nominal test for the governing coefficient is inflated roughly sevenfold. Persistent physiological series counterfeit both bimodality and bistability, so calibration against unit-root surrogates is not optional. The bimodality result in Sec. V is the cautionary case in miniature: a marginal distribution decisively bimodal against a random-walk null, accompanied by dynamics containing no cubic term at all. A study reporting only the distributional evidence would have concluded the opposite of the correct answer.

The second is about the estimator, and is the one we did not anticipate. Rolling-window autocorrelation and variance are usually treated as a transparent window onto critical slowing down — noisy, perhaps, but directionally trustworthy. Section V-F shows they are not, on a system built to contain the phenomenon: they return exponents of the wrong sign, and they do so for a structural reason rather than through sampling noise, so more data does not fix it. Increasing n from 1500 to 20,000 does not recover the sign.

This suggests a specific methodological reorientation. Instead of measuring indicators derived from an assumed local equilibrium, fit the dynamical model and test its

parameters — which is what our θ_1 test does, and which is why that test behaves correctly under calibration while the indicator-based test does not. A model-based test can be validated on ground truth before it is believed. A rolling-window indicator usually is not.

VII. Limitations and Threats to Validity

Population. No autistic participants are analysed. The mechanism is demonstrated on stress and workload corpora; transfer is an assumption, specified and pre-registered in Sec. VIII, not a finding.

Proxy validity. Tonic electrodermal activity is a proxy for a latent construct, not the construct. It is sensitive to temperature, hydration, electrode drift and movement. We mitigate with within-subject robust normalisation, motion covariates and a disjoint-sensor design, and we report a second configuration and a negative control; none of this makes the proxy the thing itself.

Persistence. The unit-root confound is intrinsic and cannot be fully eliminated, only bounded. Our surrogate test bounds it; it does not remove it.

Identifiability of the slow variable. Parameter recovery is excellent for eight of nine parameters and weak for ε ($r \approx 0.51$ in simulation). Conclusions resting on ε — principally P4 — are correspondingly weaker and are reported as exploratory.

Dimensionality. A one-dimensional load coordinate is a strong simplification; sensory and executive load may occupy separate dimensions with their own thresholds. The cusp is the minimal geometry producing the phenomena of interest, not a claim that the true system is one-dimensional.

Labels. D2 and D3 lack momentary ground truth at window resolution, so state validation rests mainly on D1, which is the corpus least suited to the early-warning analysis. That asymmetry is unavoidable with currently available open data.

What a null cannot establish. Absence of evidence is bounded evidence of absence, and the bound is only as good as the effect size assumed in the power calculation. Our power analysis covers cusp dynamics with $\lambda \geq 0.05$ and $\alpha_0 \geq 0.6$ in the units of a within-subject standardised load coordinate. A genuine cusp weaker than that, or one whose control parameters are driven by variables we did not measure, would not have been detected. We have specified the detectable region rather than claiming to have excluded every version of the hypothesis.

Multiplicity and unit definition. Nurse sessions are numerous and of uneven length, and 102 of the 147 units come from 15 nurses, so sessions within a nurse are not independent. Treating each session as a unit gives that corpus most of the weight in any pooled proportion and understates the standard error. We therefore report per-corpus figures alongside pooled ones, and recompute the headline rate with one vote per person and a cluster-robust interval (Sec. efsec:results); the conclusion strengthens

rather than weakens under that reweighting. No conclusion in the paper depends on a threshold being crossed in the pooled figure alone.

Scope of the estimator critique. Section V-F tests fixed-width rolling autocorrelation and variance, with and without Gaussian detrending, which is what the applied literature predominantly uses. It does not test adaptive-bandwidth estimators, spectral indicators such as the spectral reddening ratio, or generalised-least-squares approaches to the same quantity. We claim the failure for the estimators we tested and do not claim it for the family in general. Nor do we claim it for every dynamical setting: our simulation has moving control parameters and a slow variable, which is a harder case than the fixed-parameter fold usually studied. Whether the same censoring occurs under quasi-static forcing is a question this paper raises rather than settles.

VIII. Toward Validation in Autistic Adults

The null reported above is uninformative about autistic adults, because it was obtained in populations where the mechanism is not hypothesised to operate. We therefore specify, in advance, the study that would test it, and what result would end the line of enquiry.

Design. 28 days of ecological momentary assessment (5 prompts/day: sensory load, task-switching demand, uncertainty, perceived control) with continuous wrist EDA, BVP, TEMP and ACC, in autistic adults and a matched non-autistic comparison group, sampling both occupational and domestic settings [9]. Two time series per participant — physiological at seconds, self-report at hours — because the analysis here cannot rule out that the relevant dynamics live on the slower one.

Primary hypothesis. The fitted splitting factor α_0 is larger in the autistic group, so a second attractor exists at lower demand. This is a between-group comparison of a parameter, not a within-group detection, and is therefore better powered than what was attempted here.

Secondary. P1–P4 hold within the autistic group; α_A correlates with self-reported recovery time; the group difference in α_0 survives adjustment for mean arousal, so that it is not simply a stress-level effect.

Analysis, fixed in advance. The estimator, the $\theta_1 \geq 0$ constraint, the retention of non-identified units, and surrogate-calibrated inference are all as specified in Sec. IV-C–IV-E. Nominal p -values will not be used.

Falsification. The framework is rejected if the hysteresis exponent excludes $3/2$, if the early-warning exponents exclude $\mp 1/2$, or if the unit-root surrogate ensemble reproduces the observed statistics — the same criteria applied here, applied to the population the model is about. A second well-powered null under those conditions should be read as evidence that sensory–executive overload is not a cusp catastrophe at physiological timescales, and we commit to reporting it as such.

Ethics. Autistic-led advisory involvement in protocol design, burden minimisation, and explicit consent for passive sensing.

IX. Conclusion

Modelling functional overload as motion in a cusp potential replaces a descriptive state-transition account with a generative one whose consequences are computable in advance. Discrete states become metastable basins, the transition kernel becomes a function of nine interpretable parameters, abrupt onset and delayed recovery become geometric necessities, and two scaling laws — $a^{3/2}$ for the hysteresis loop, $\mp 1/2$ for the early-warning exponents — become point hypotheses that data can reject.

Tested against those hypotheses on 147 recordings from 40 people across three open wearable corpora, only one of the five proves evaluable, and it is not supported. Calibrated power of 1.00 for strong and 0.76 for moderate cusp dynamics at the observed series lengths excludes those regimes; power of 0.24 in the weak regime does not exclude a faint fold, and we say so rather than round the claim upward. The population, proxy and timescale caveats of Sec. V apply on top of that. Along the way we measured something we did not set out to measure: on near-unit-root physiological series the standard significance test for the governing coefficient has a false-positive rate of 0.42 rather than 0.05, and the rolling-window early-warning indicators return exponents of the wrong sign even on data containing a genuine fold. Since claims of bistability and critical transition in wearable physiology are routinely made with uncalibrated tests, we expect that multiplier to be the more consequential of our two findings.

The construct the model was built for — sensory-executive overload in autistic adults — remains unmeasured, because no public dataset of the required density exists. We have said so rather than titled the paper after it, and specified in advance what such a dataset would have to show for the model to survive.

Code and Data Availability

The estimator, the analysis pipeline, the Monte Carlo suite, the figure scripts and this manuscript are released under the MIT licence at <https://github.com/n-3-0-l-d-3-v/overload-cusp-catastrophe>, together with a provenance table recording the replication count behind every number reported. The three corpora are third-party open datasets and are not redistributed: WESAD from the UCI Machine Learning Repository (ID 465), and Wearable Exam Stress and Nurse Stress from PhysioNet. Each was obtained programmatically from its primary repository under that repository’s own terms, with the exact URLs, download dates and byte counts recorded in the artefact.

Acknowledgment

Analyses use the WESAD [38], Wearable Exam Stress [39] and Nurse Stress [40] datasets; we thank their authors

for open release. An AI assistant was used mainly for repeated model test runs, including the simulation and validation runs. The model, the analysis and the final manuscript are the work of the authors, who verified all results and are solely responsible for the content.

References

- [1] K. MacLennan, S. O’Brien, and T. Tavassoli, “In our own words: The complex sensory experiences of autistic adults,” *Journal of Autism and Developmental Disorders*, vol. 52, no. 7, pp. 3061–3075, 2022.
- [2] M. Kiep, A. Spek, E. Ceulemans, and I. Noens, “Sensory processing and executive functioning in autistic adults,” *Journal of Autism and Developmental Disorders*, vol. 55, no. 6, pp. 2075–2084, 2025.
- [3] E. A. Demetriou, A. Lampit, D. S. Quintana, S. L. Naismith, Y. J. C. Song, J. E. Pye, I. Hickie, and A. J. Guastella, “Autism spectrum disorders: A meta-analysis of executive function,” *Molecular Psychiatry*, vol. 23, no. 5, pp. 1198–1204, 2018.
- [4] E. A. Demetriou, M. M. DeMayo, and A. J. Guastella, “Executive function in autism spectrum disorder: History, theoretical models, empirical findings, and potential as an endophenotype,” *Frontiers in Psychiatry*, vol. 10, p. 753, 2019.
- [5] E. Pellicano and D. Burr, “When the world becomes ‘too real’: A Bayesian explanation of autistic perception,” *Trends in Cognitive Sciences*, vol. 16, no. 10, pp. 504–510, 2012.
- [6] N. Angeletos Chrysaitis and P. Seriès, “10 years of Bayesian theories of autism: A comprehensive review,” *Neuroscience & Biobehavioral Reviews*, vol. 145, p. 105022, 2023.
- [7] M. Adamou, A. Kehagias, and G. Antoniou, “A multi-scale mathematical framework for modelling the dynamic nature of autism spectrum disorder symptoms: Integrating predictive coding, information theory, and network principles,” *Frontiers in Psychiatry*, vol. 17, p. 1787120, 2026.
- [8] S. Cano, C. Cubillos, R. Alfaro, and A. Romo, “Wearable solutions using physiological signals for stress monitoring on individuals with autism spectrum disorder (ASD): A systematic literature review,” *Sensors*, vol. 24, no. 24, p. 8137, 2024.
- [9] Y. Chen, Z. Xi, T. Greene, and W. Mandy, “A systematic review of ecological momentary assessment in autism research,” *Autism*, vol. 29, no. 6, pp. 1374–1389, 2025.
- [10] A. M. Scheeren, S. Nieuwenhuis, L. Crane, Y. Roke, and S. Begeer, “Masking, social context and perceived stress in autistic adults: An ecological momentary assessment study,” *Autism*, vol. 29, no. 12, pp. 3002–3013, 2025.
- [11] B. S. Siepe, J. M. B. Haslbeck, M. Kloft, and A. Büchner, “Introducing openESM: A database of openly available experience sampling datasets,” *Behavior Research Methods*, vol. 58, p. 240, 2026.
- [12] P. Aug’e, A. Maruani, E. Humeau, P. Ellul, A. Cartigny, A. Lefebvre, F. Dellapiazza, and R. Delorme, “Global sensory features are linked to executive and attentional impairments in autism spectrum disorders,” *Journal of Autism and Developmental Disorders*, vol. 55, no. 8, pp. 2788–2796, 2025.
- [13] T. Arthur, S. Vine, G. Buckingham, M. Brosnan, M. Wilson, and D. Harris, “Testing predictive coding theories of autism spectrum disorder using models of active inference,” *PLOS Computational Biology*, vol. 19, no. 9, p. e1011473, 2023.
- [14] R. Thom, *Structural Stability and Morphogenesis*. Reading, MA, USA: W. A. Benjamin, 1975.
- [15] R. Gilmore, *Catastrophe Theory for Scientists and Engineers*. New York, NY, USA: Wiley, 1981.
- [16] E. C. Zeeman, “Catastrophe theory,” *Scientific American*, vol. 234, no. 4, pp. 65–83, 1976.
- [17] H. L. J. van der Maas and P. C. M. Molenaar, “Stagewise cognitive development: An application of catastrophe theory,” *Psychological Review*, vol. 99, no. 3, pp. 395–417, 1992.
- [18] R. P. P. P. Grasman, H. L. J. van der Maas, and E.-J. Wagenmakers, “Fitting the cusp catastrophe in R: A cusp package primer,” *Journal of Statistical Software*, vol. 32, no. 8, pp. 1–27, 2009.

- [19] X. Li, T. Henning, and C. Camerer, "Estimating hidden Markov models (HMMs) of the cognitive process in strategic thinking using eye-tracking," *Frontiers in Behavioral Economics*, vol. 2, p. 1225856, 2023.
- [20] I. W. Eisenberg, P. G. Bissett, A. Z. Enkavi, J. Li, D. P. MacKinnon, L. A. Marsch, and R. A. Poldrack, "Uncovering the structure of self-regulation through data-driven ontology discovery," *Nature Communications*, vol. 10, p. 2319, 2019.
- [21] R. S. Zahler and H. J. Sussmann, "Claims and accomplishments of applied catastrophe theory," *Nature*, vol. 269, pp. 759–763, 1977.
- [22] H. J. Sussmann and R. S. Zahler, "Catastrophe theory as applied to the social and biological sciences: A critique," *Synthese*, vol. 37, no. 2, pp. 117–216, 1978.
- [23] C. Wissel, "A universal law of the characteristic return time near thresholds," *Oecologia*, vol. 65, no. 1, pp. 101–107, 1984.
- [24] M. Scheffer, J. Bascompte, W. A. Brock, V. Brovkin, S. R. Carpenter, V. Dakos, H. Held, E. H. van Nes, M. Rietkerk, and G. Sugihara, "Early-warning signals for critical transitions," *Nature*, vol. 461, pp. 53–59, 2009.
- [25] M. Scheffer, S. R. Carpenter, T. M. Lenton, J. Bascompte, W. Brock, V. Dakos, J. van de Koppel, I. A. van de Leemput, S. A. Levin, E. H. van Nes, M. Pascual, and J. Vandermeer, "Anticipating critical transitions," *Science*, vol. 338, no. 6105, pp. 344–348, 2012.
- [26] V. Dakos, S. R. Carpenter, W. A. Brock, A. M. Ellison, V. Guttal, A. R. Ives, S. K'efi, V. Livina, D. A. Seekell, E. H. van Nes, and M. Scheffer, "Methods for detecting early warnings of critical transitions in time series illustrated using simulated ecological data," *PLOS ONE*, vol. 7, no. 7, p. e41010, 2012.
- [27] I. A. van de Leemput, M. Wichers, A. O. J. Cramer, D. Borsboom, F. Tuerlinckx, P. Kuppens, E. H. van Nes, W. Viechtbauer, E. J. Giltay, S. H. Aggen, C. Derom, N. Jacobs, K. S. Kendler, H. L. J. van der Maas, M. C. Neale, F. Peeters, E. Thiery, P. Zachar, and M. Scheffer, "Critical slowing down as early warning for the onset and termination of depression," *Proceedings of the National Academy of Sciences USA*, vol. 111, no. 1, pp. 87–92, 2014.
- [28] M. Wichers, A. C. Smit, and E. Snippe, "Early warning signals based on momentary affect dynamics can expose nearby transitions in depression: A confirmatory single-subject time-series study," *Journal for Person-Oriented Research*, vol. 6, no. 1, pp. 1–15, 2020.
- [29] A. C. Smit, M. A. Helmich, L. F. Bringmann, A. J. Oldehinkel, M. Wichers, and E. Snippe, "Critical slowing down in momentary affect as early warning signal of impending transitions in depression," *Clinical Psychological Science*, vol. 13, no. 4, pp. 760–773, 2025.
- [30] M. A. Helmich, M. Olthof, A. J. Oldehinkel, M. Wichers, L. F. Bringmann, and A. C. Smit, "Early warning signals and critical transitions in psychopathology: Challenges and recommendations," *Current Opinion in Psychology*, vol. 41, pp. 51–58, 2021.
- [31] P. D. Ditlevsen and S. J. Johnsen, "Tipping points: Early warning and wishful thinking," *Geophysical Research Letters*, vol. 37, no. 19, p. L19703, 2010.
- [32] C. Boettiger and A. Hastings, "Quantifying limits to detection of early warning for critical transitions," *Journal of the Royal Society Interface*, vol. 9, no. 75, pp. 2527–2539, 2012.
- [33] W. Boucsein, *Electrodermal Activity*, 2nd ed. New York, NY, USA: Springer, 2012.
- [34] M. Benedek and C. Kaernbach, "A continuous measure of phasic electrodermal activity," *Journal of Neuroscience Methods*, vol. 190, no. 1, pp. 80–91, 2010.
- [35] A. Greco, G. Valenza, A. Lanata, E. P. Scilingo, and L. Citi, "cvxEDA: A convex optimization approach to electrodermal activity processing," *IEEE Transactions on Biomedical Engineering*, vol. 63, no. 4, pp. 797–804, 2016.
- [36] H. F. Posada-Quintero and K. H. Chon, "Innovations in electrodermal activity data collection and signal processing: A systematic review," *Sensors*, vol. 20, no. 2, p. 479, 2020.
- [37] F. Shaffer and J. P. Ginsberg, "An overview of heart rate variability metrics and norms," *Frontiers in Public Health*, vol. 5, p. 258, 2017.
- [38] P. Schmidt, A. Reiss, R. Duerichen, C. Marberger, and K. Van Laerhoven, "Introducing WESAD, a multimodal dataset for wearable stress and affect detection," in *Proc. 20th ACM Int. Conf. Multimodal Interaction (ICMI)*, 2018, pp. 400–408.
- [39] M. R. Amin, D. Wickramasuriya, and R. T. Faghih, "A wearable exam stress dataset for predicting cognitive performance in real-world settings (version 1.0.0)," *PhysioNet*, 2022.
- [40] S. Hosseini, R. Gottumukkala, S. Katragadda, R. T. Bhupatiraju, Z. Ashkar, C. W. Borst, and K. Cochran, "A multimodal sensor dataset for continuous stress detection of nurses in a hospital," *Scientific Data*, vol. 9, p. 255, 2022.
- [41] H. A. Kramers, "Brownian motion in a field of force and the diffusion model of chemical reactions," *Physica*, vol. 7, no. 4, pp. 284–304, 1940.
- [42] C. W. Gardiner, *Stochastic Methods: A Handbook for the Natural and Social Sciences*, 4th ed. Berlin, Germany: Springer, 2009.
- [43] M. Hutzenthaler, A. Jentzen, and P. E. Kloeden, "Strong convergence of an explicit numerical method for SDEs with non-globally Lipschitz continuous coefficients," *Annals of Applied Probability*, vol. 22, no. 4, pp. 1611–1641, 2012.
- [44] S. J. Julier and J. K. Uhlmann, "Unscented filtering and nonlinear estimation," *Proceedings of the IEEE*, vol. 92, no. 3, pp. 401–422, 2004.
- [45] T. Schreiber and A. Schmitz, "Improved surrogate data for nonlinearity tests," *Physical Review Letters*, vol. 77, no. 4, pp. 635–638, 1996.
- [46] I. Nahum-Shani, S. N. Smith, B. J. Spring, L. M. Collins, K. Witkiewitz, A. Tewari, and S. A. Murphy, "Just-in-time adaptive interventions (JITAIs) in mobile health: Key components and design principles for ongoing health behavior support," *Annals of Behavioral Medicine*, vol. 52, no. 6, pp. 446–462, 2018.